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Tailorable Force-Sensing Skins for Manipulators

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Introduction

Two removable tactile skins extend sensing beyond the gripper: a textile sleeve for low-force contact and a conformal printed shell for higher-threshold interaction.

Methods

Shared design: Velostat taxel matrices, 20 mm pitch, and under $10 in materials per link (readout excluded).

Textile sleeve: Velostat between conductive-fabric electrodes detects light contact.

Printed shell: copper electrodes inside 95A TPU; a normally open gap raises the detection threshold.

Testing: four 0–30 N normal load–unload cycles per taxel (Instron).

Results

Operating range: ~0–6 N textile; ~4–12 N AM, for the tested configurations.

Repeatability: mean multi-force resistance CV of 0.052 (textile) and 0.066 (AM); lower is better. AM has greater hysteresis.

Robot demos: textile sensing supports contact avoidance; AM enables arm contact to open a sliding door while holding a dustpan.

Acknowledgment

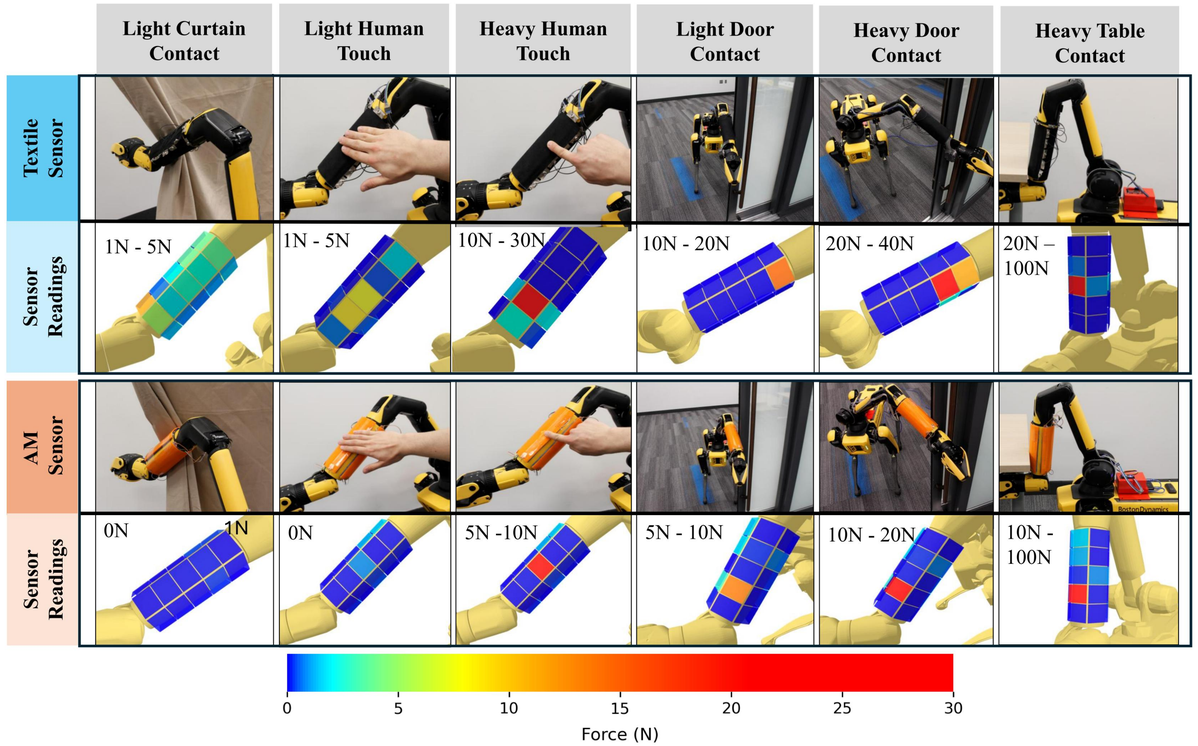
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Robotics: Perception & Manipulation Lab

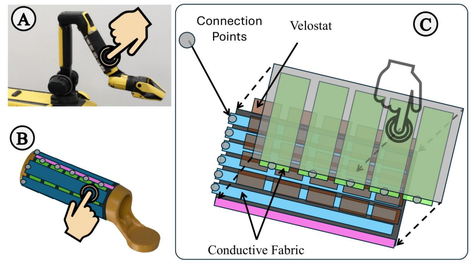
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Low-cost skins for avoiding and embracing whole-arm contact.

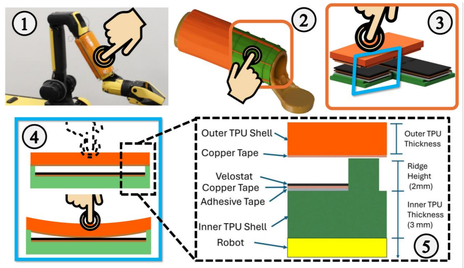
Qualitative Results



Textile Sleeve



Additive Manufacturing (AM)



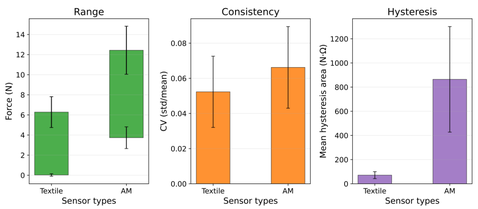
Contact Avoidance



Contact Embracing



Characterization



Range: Detection threshold to saturation
Consistency: Reading variation (lower is better)
Hysteresis: Loading–unloading response difference